

Lecture 25

- Complete repetition code for bit flips.
 - stabilizer operators
 - error syndrome measurements.
- repetition code for phase flip errors
- Shor code for arbitrary errors

$$|0_L\rangle = |000\rangle, |1_L\rangle = |111\rangle$$

$$X_L = |000\rangle\langle 111| + |111\rangle\langle 000|$$

$$= X_2 X_1 X_0 \text{ or } X \otimes X \otimes X, \text{ where } X_2 = X \otimes I \otimes I.$$

$$Z_L = Z_2 Z_1 Z_0 \text{ or } Z_2 \text{ or } Z_1 \text{ or } Z_0$$

Code distance $d=3$ for X errors

$d=1$ for Z errors

it's as if you don't have an error correction at all.

$$Y_L = i X_L Z_L$$

Types of Errors

$$\left\{ \begin{array}{l} X \\ Y \\ Z \\ e^{i\frac{\theta}{2}X} \end{array} \right.$$

coherent error

$$e^{i\frac{\theta}{2}X} = \cos\frac{\theta}{2} I - i \sin\frac{\theta}{2} X$$

no error bit flip error

$$\sigma^+ = |0\rangle\langle 1| \quad \text{decay error}$$

Focus on X errors

4 focus

$$\left\{ \begin{array}{l} I \\ X_0 \\ X_1 \\ X_2 \end{array} \right.$$

$$|\psi\rangle = \alpha|000\rangle + \beta|111\rangle$$

$$X_0|\psi\rangle = \alpha|001\rangle + \beta|110\rangle$$

how do you correct
 $\rightarrow$ Measure & collapse.
 however, I want to preserve α & β .

we only need 2 bits of classical info to distinguish 4 error states.

Stabilizer $\Rightarrow$ sum that $S_0|\psi\rangle = +1 \cdot |\psi\rangle \quad \forall \alpha, \beta.$

$$S_0 = Z_1 Z_0 = I \otimes Z \otimes Z$$

$\rightarrow$ detects error in q_1, q_0

$\uparrow$ always ± 1 eigenvalue.

$$S_1 = Z_2 Z_1 = Z \otimes Z \otimes I$$

$\rightarrow$ detects error in q_2, q_1

$$S_2 = S_0 S_1 = Z \otimes I \otimes Z$$

$[S_0, S_1] = 0 \rightarrow$ doesn't matter which order you observe them.

neither affects the other.

$$[S_j, P_L] = 0$$

$j=0,1$ logical Pauli

$$(Z \otimes I \otimes Z) \otimes (Z \otimes I \otimes Z) \otimes (I \otimes X \otimes Z) - (I \otimes X \otimes Z) \otimes (Z \otimes I \otimes Z) \otimes (Z \otimes I \otimes Z)$$

$\rightarrow$ contributes 2 negative signs.

Measuring the stabilizer gives you the info:

$\hookrightarrow$ convince yourself that

$$\langle \psi | S_0 | \psi \rangle = \langle \psi | I \otimes Z \otimes Z [\alpha | 001 \rangle + \beta | 110 \rangle]$$

$$= \langle \psi | [-\alpha | 001 \rangle - \beta | 110 \rangle]$$

$$= [\alpha \langle 001 | + \beta \langle 110 |] [-\alpha | 001 \rangle - \beta | 110 \rangle]$$

$$= -|\alpha|^2 - |\beta|^2$$

$$S_0(X_0|\psi\rangle) = -X_0|\psi\rangle$$

$$S_1(X_1|\psi\rangle) = +X_1|\psi\rangle$$

how can you measure an operator in practice.
 $\hookrightarrow$ not destroy the state.

Sometimes stabilizes anticommute with one type of error,

$$S_0 X_0 = Z_1 Z_0 X_0$$

$$X_0 S_0 = X_0 Z_0 Z_1 = -S_0 X_0$$

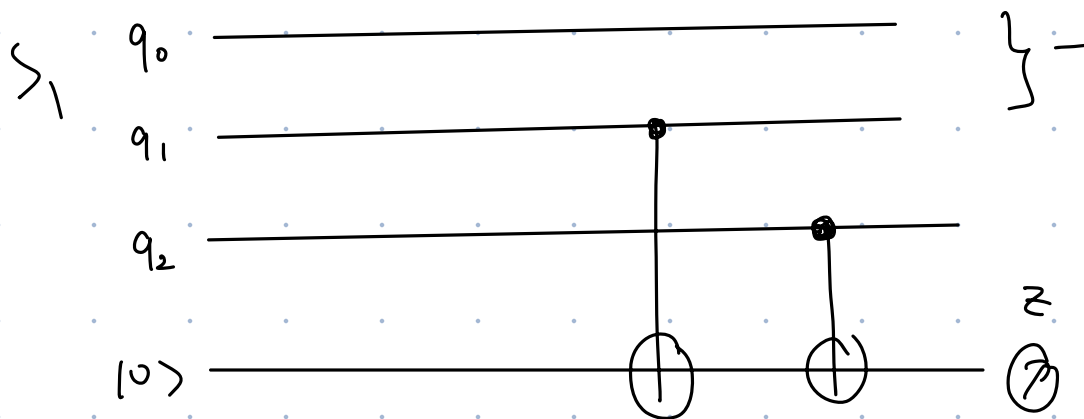
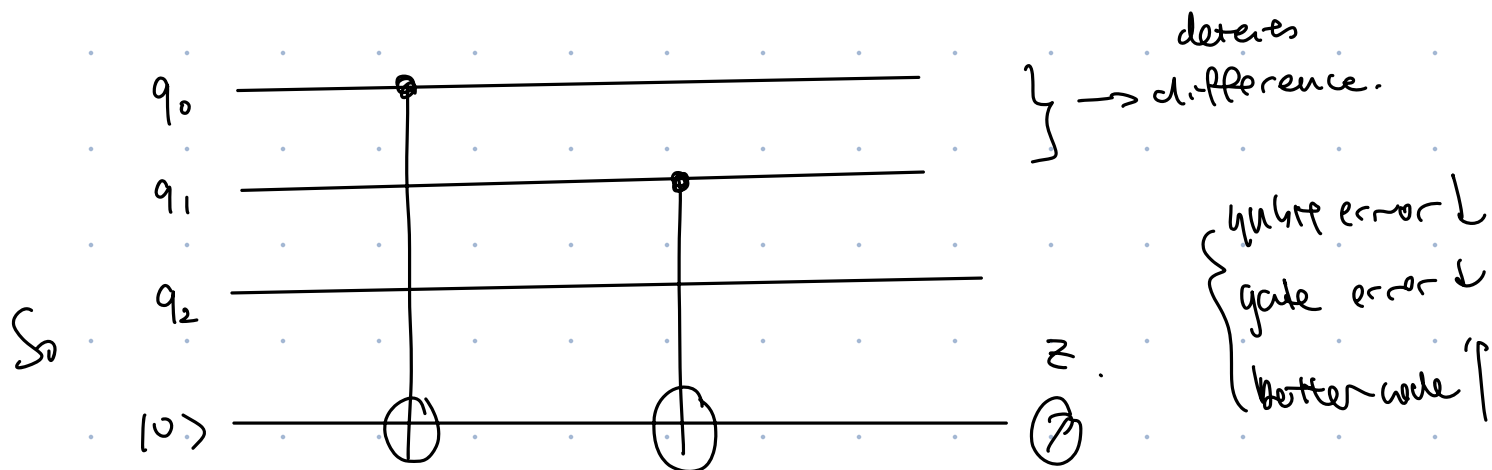
Error	$S_1 = Z_2 Z_1$	$S_0 = Z_1 Z_0$	Correction apply
I	+1	+1	nothing
X_0	+1	-1	
X_1	-1	-1	
X_2	-1	+1	

Is there any advantage to encoding information in qubits?



how can you measure an operator in practice.
& not destroy the state.

→ Measurement Results are called
"error syndrome."



Phase-flip repetition code

$$|0_L\rangle = H^{\otimes 3} |000\rangle = |+++ \rangle$$

$$|1_L\rangle = H^{\otimes 3} |111\rangle = |--\rangle$$

$$X_L = H^{\otimes 3} X^{\otimes 3} H^{\otimes 3} = Z_2 Z_1 Z_0$$

$$Z_L = X_2 X_1 X_0 \equiv \text{or } X_2 \text{ or } X_1 \text{ or } X_0$$

$$Z|+\rangle = |-\rangle, Z|-\rangle = |+\rangle.$$

the bit flip operator in the conjugate basis.

$$S_0 = H^{\otimes 3} Z_1 Z_0 H^{\otimes 3} = X_1 X_0$$

$$S_1 = X_2 X_1$$

Suppose phase flip error Z_1

$$|4\rangle = \alpha |+++ \rangle + \beta |--\rangle$$

$$Z_1 |4\rangle = \alpha |+-+\rangle + \beta |-+-\rangle$$

$$S_0 = -1$$

$$S_1 = -1$$

How do you make Hadamard

Shor:

(i) construct phase flips:

$$|0_L\rangle = |+\rangle^{\otimes 3}$$

$$|1_L\rangle = |-\rangle^{\otimes 3}$$

(ii) ~~Construct~~ Replace

$$|+\rangle \rightarrow \frac{1}{\sqrt{2}} (|000\rangle + |111\rangle)$$

$$|-\rangle \rightarrow \frac{1}{\sqrt{2}} (|000\rangle - |111\rangle)$$

(iii)

$$|0_L\rangle = \frac{1}{\sqrt{8}} (|000\rangle + |111\rangle) (|000\rangle + |111\rangle) (|000\rangle + |111\rangle)$$